Multimode emission and hysteresis in FIB-engineered random laser diodes

Andrea Martínez Pacheco[1], Sara Rodríguez Corvillo[2], Alberto Rivera-Calzada[2], **Antonio Consoli**[1,3], Jacobo Santamaría[2] and Cefe López[1]

[1]Instituto de Ciencia de Materiales de Madrid (ICMM), Consejo Superior de Investigaciones Científicas (CSIC) Madrid, Spain
[2]GFMC, Departamento de Física de Materiales, Facultad de Física, Universidad Complutense, Madrid, Spain
[3]Escuela de Ingeniería de Fuenlabrada (EIF), Universidad Rey Juan Carlos (URJC), Fuenlabrada, Madrid, Spain

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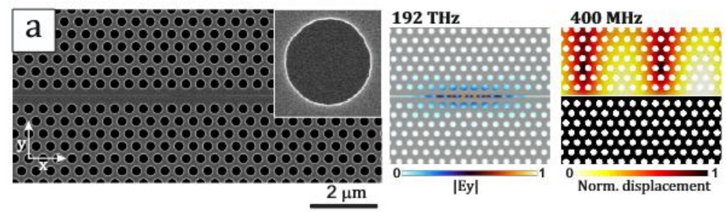
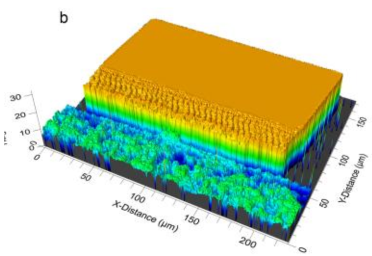
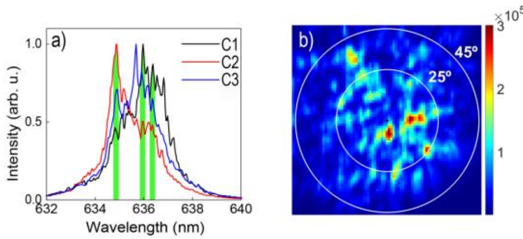
Our Team

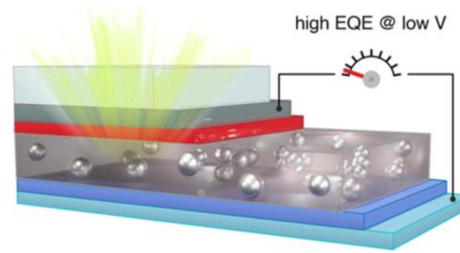
Photonics Crystal Group (ICMM-CISC)

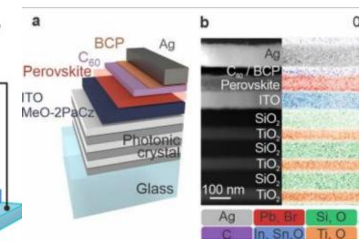


We study **disordered structures** in photonics devices, with application in **random lasing, opto-mechanics, radiative cooling, perovskite emitters.**



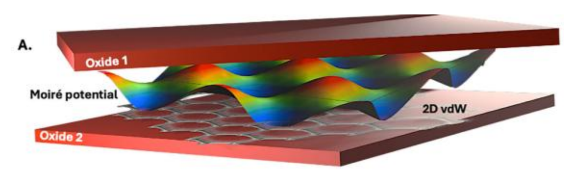


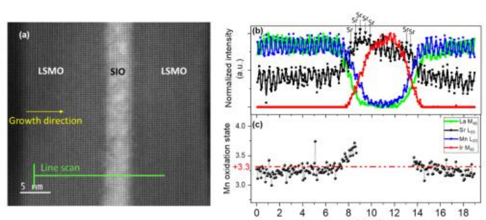


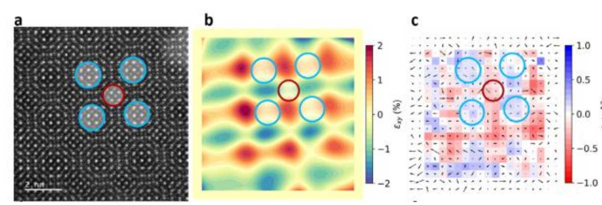


Physics of Complex Materials Group (UCM)

We are interested in proximity phenomena at interfaces between correlated oxides. Focused on magnetism, superconductivity, ferroelectricity and multiferroicity.



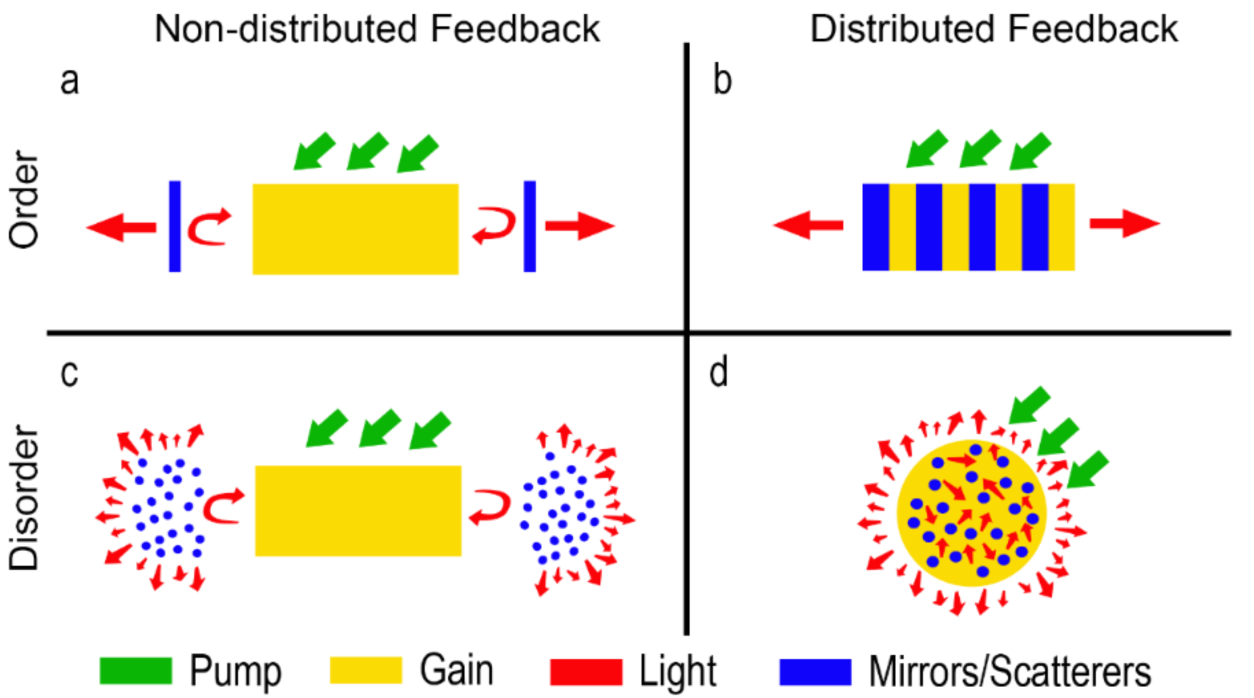




Introduction

Random Lasers

In Random Lasers, feedback is provided by scattering elements. These can be spatially distributed inside the active medium (distributed feedback) or placed at its the edges.



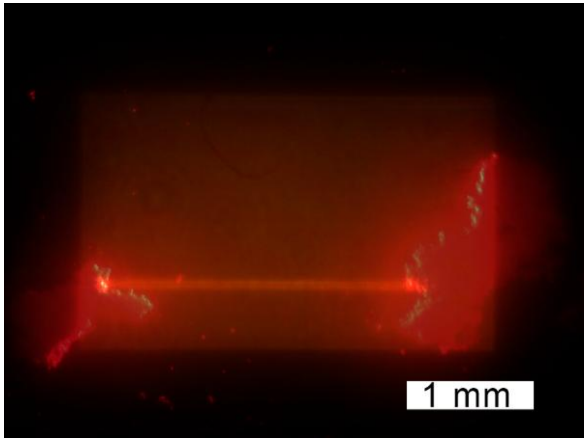
A Consoli, N Caselli, C López “Electrically driven random lasing from a modified Fabry–Pérot laser diode” Nature Photonics 16 (3), 219-225 (2022)

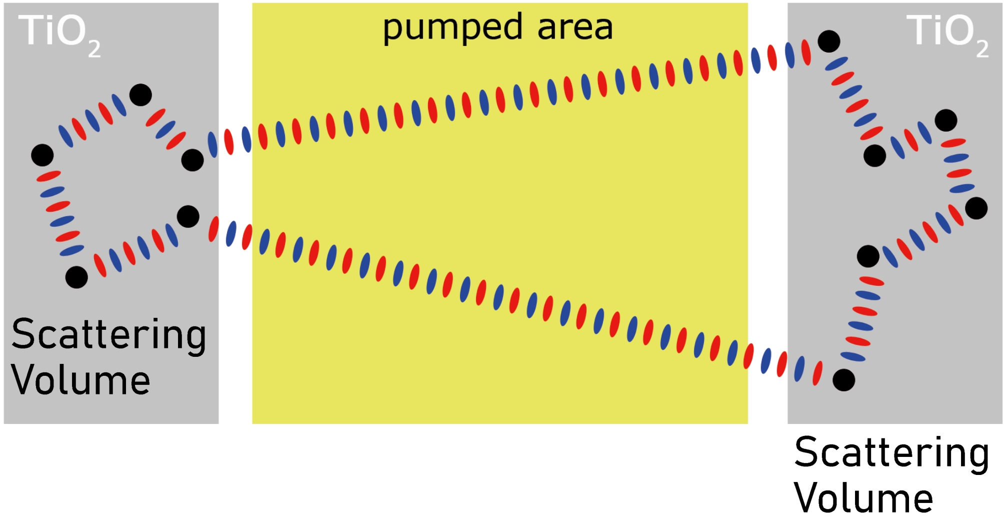
Introduction

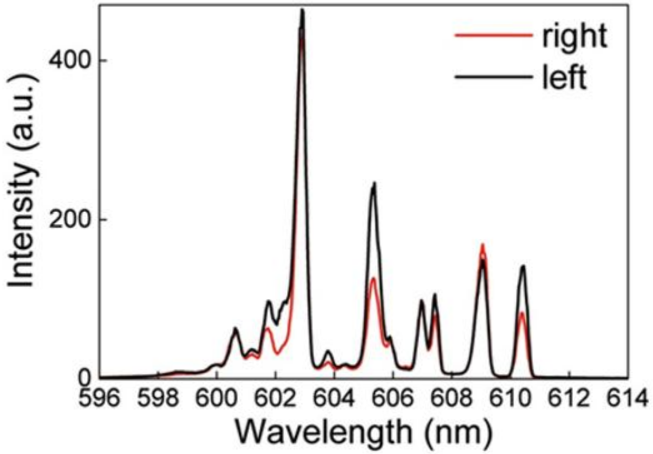
Resonances in diffusive cavity

Two scattering volumes (TiO_2 particles) enclose a liquid dye (Rhodamine)

Scattering particles adds an unknown phase to the light backscattered into the active region. If an integer number of periods sum up at a certain frequency, that frequency is allowed to exist.





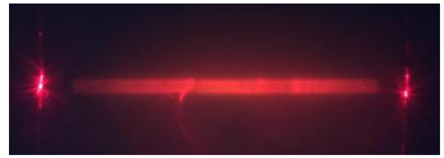


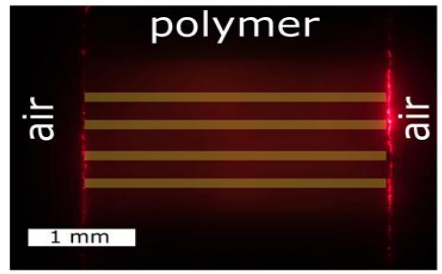
A. Consoli and C. López "Decoupling gain and feedback in coherent random lasers: experiments and simulations" Sci. Rep. 5, 16848 (2015)

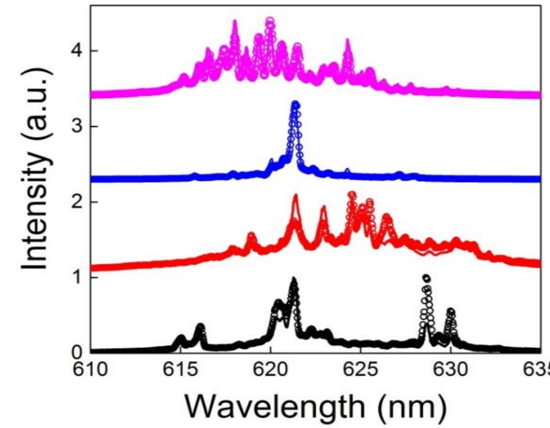
Introduction

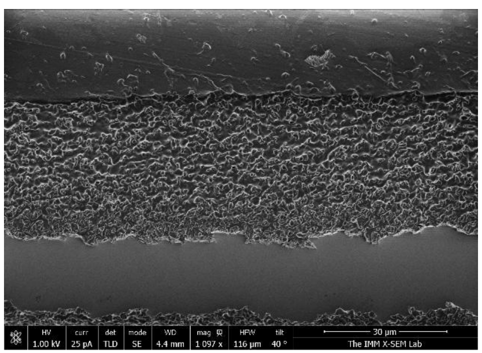
Scattering surfaces

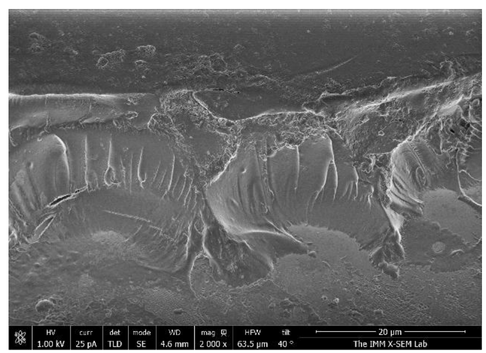
Solid dye doped polymer film with rough edges.

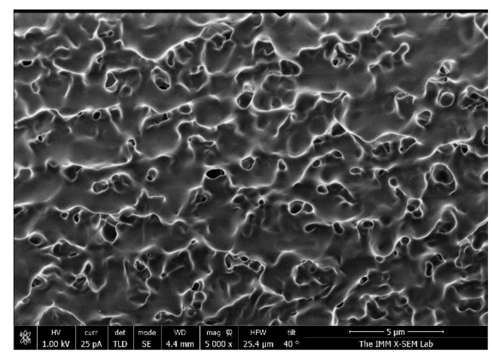








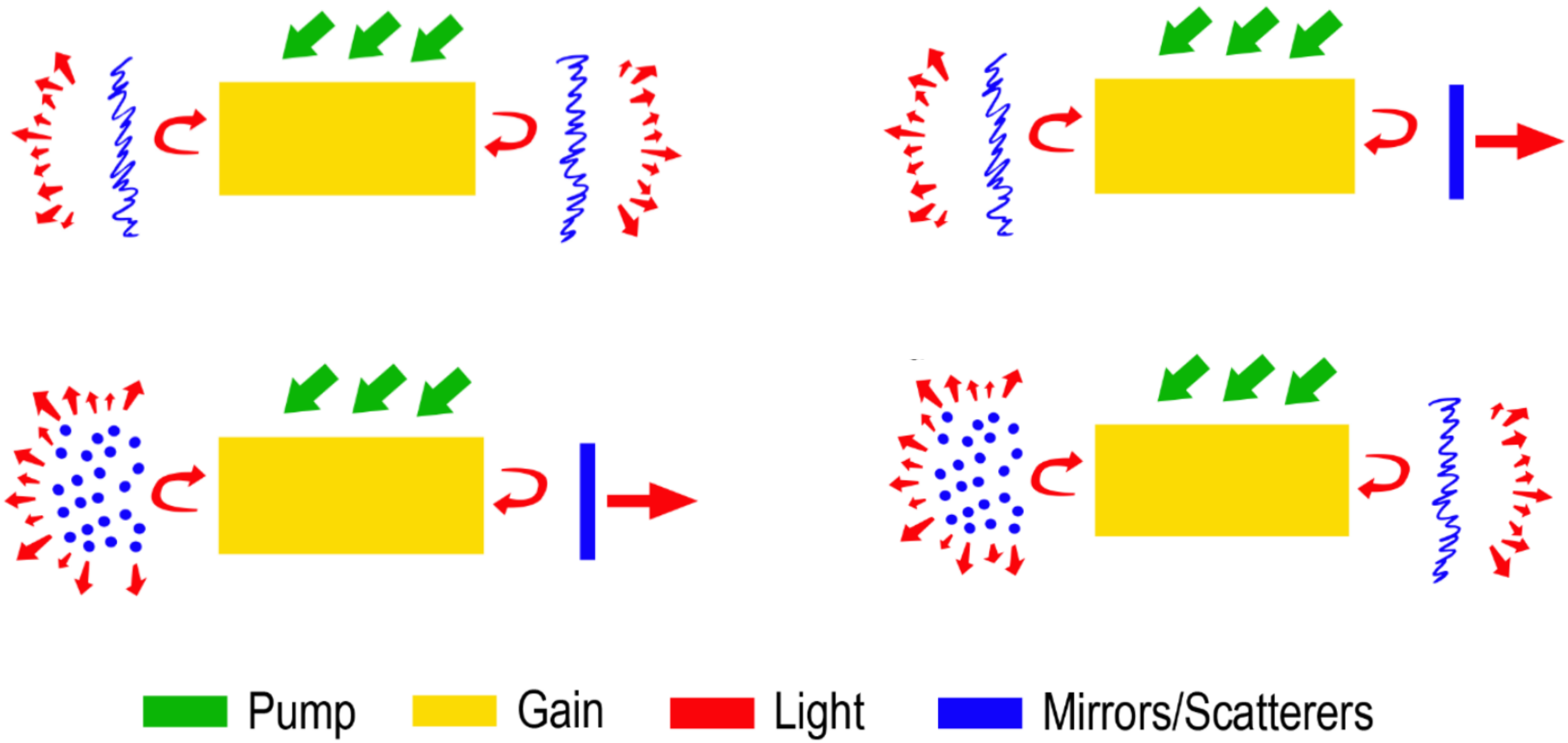




A. Consoli, D. Mariano da Silva, N.U. Wetter and C. López “Large area resonant feedback random lasers based on dye-doped biopolymer films” Opt. Express 23 (23), 29954-29963 (2015)
A. Consoli, E. Soria, N. Caselli and C. López “Random lasing emission tailored by femtosecond and picosecond pulsed polymer ablation” Optics Letters 44 (3), 518-521 (2019)

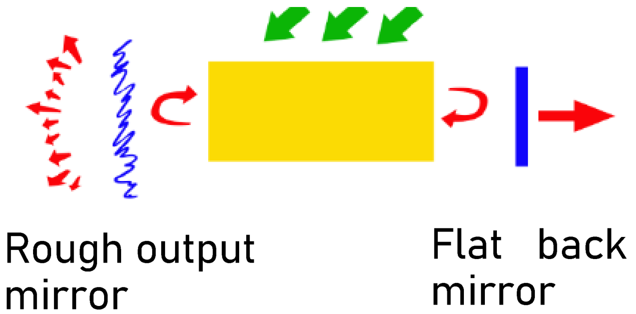
Introduction

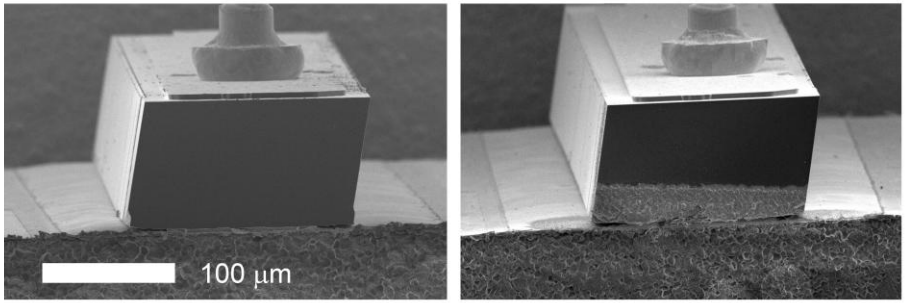
Scattering surfaces/volumes and mirrors

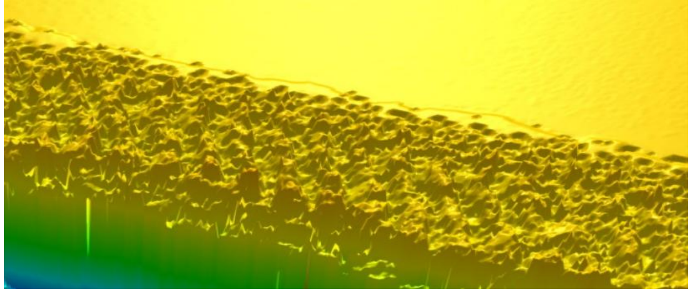


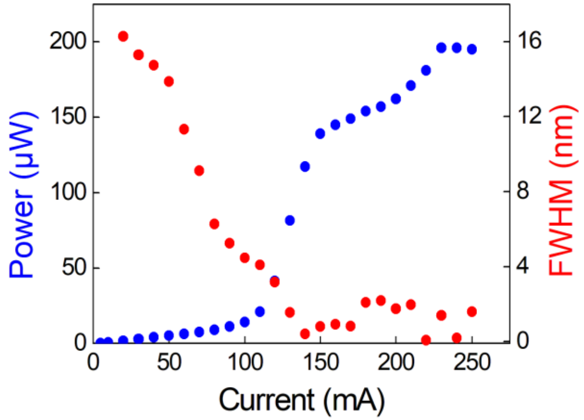
Random Laser Diodes

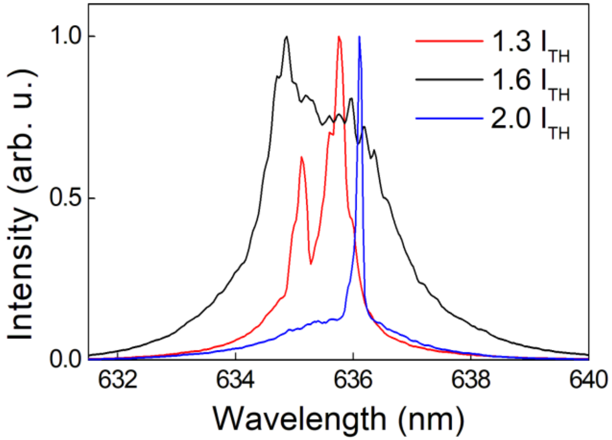
Pulse laser ablation is used to roughen the output facet of a GaAs Fabry-Perot laser diode. The modified output mirror and the untouched back mirror provide optical feedback for random lasing.

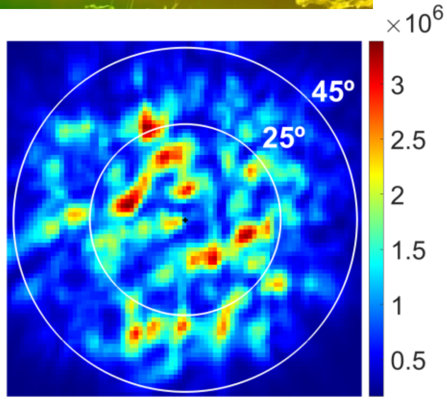










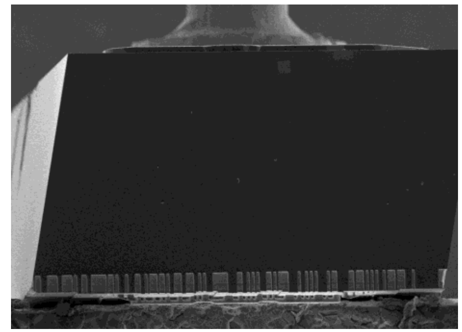


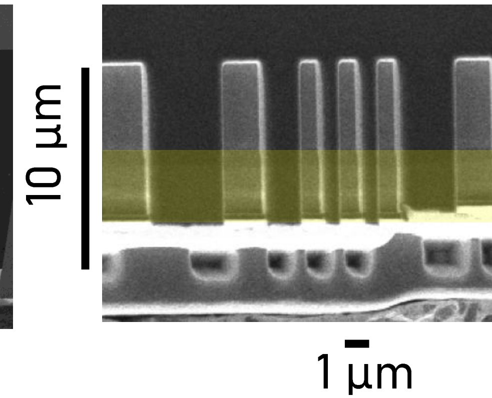
A Consoli, N Caselli, C López Nat. Photonics 16 (3), 219-225 (2022)
A Consoli, PD García, C López Opt. Express 31 (25), 42439-42448 (2023)

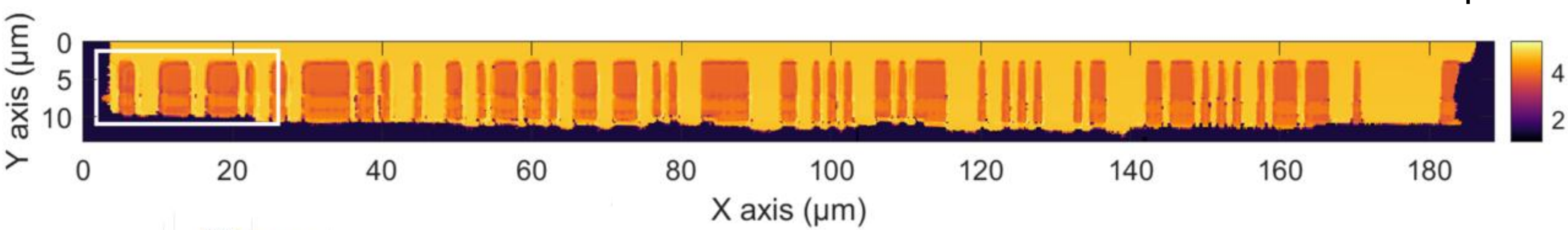
FIB-sculpted RLD

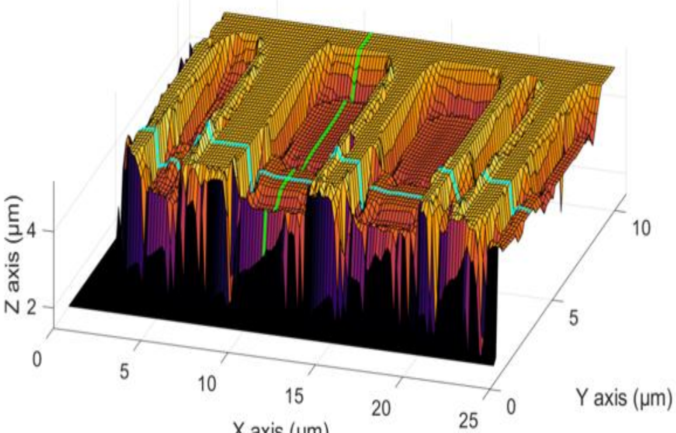
Focused Ion Beam fabrication

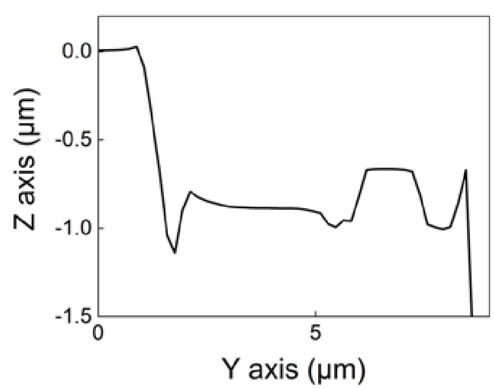
Predesigned mask, consisting of 90 rectangular slots with (1x10 um^2) randomly located along the 180 um wide emitting area.

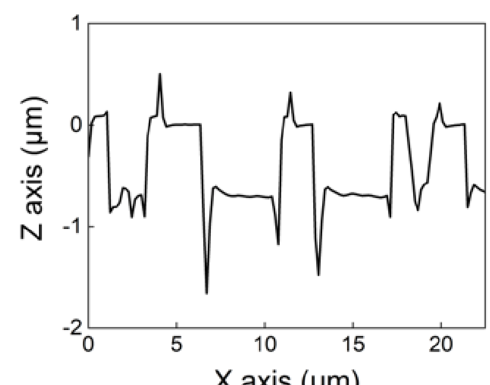






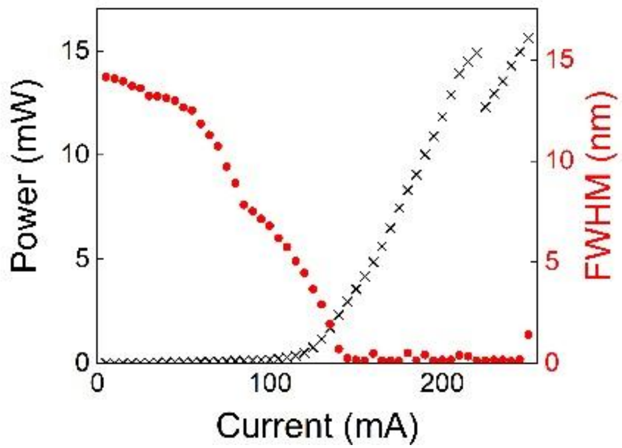


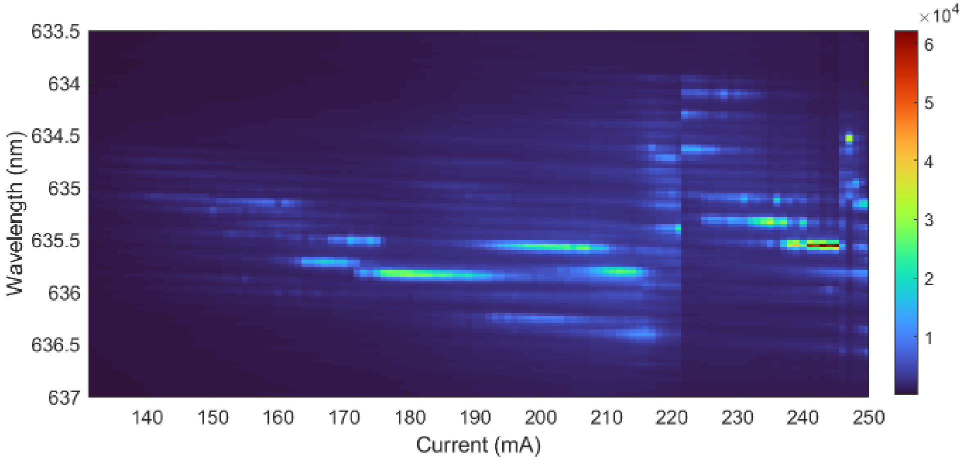


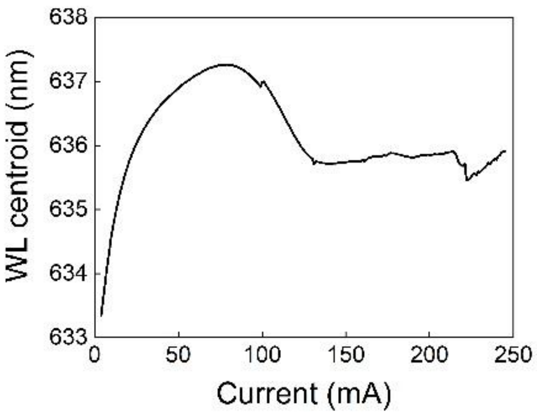


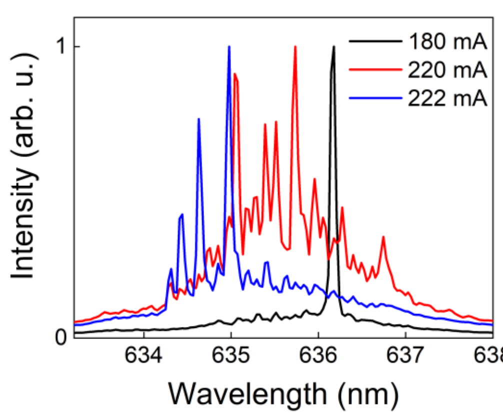
Device emission

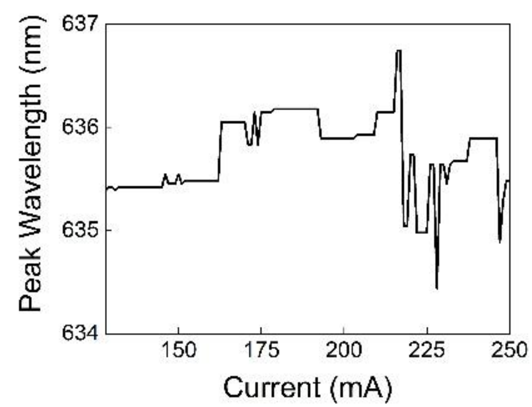
I_{TH} = 130 mA, η_{SLOPE} = 0.15 W/A, P_{MAX} = 15 mW



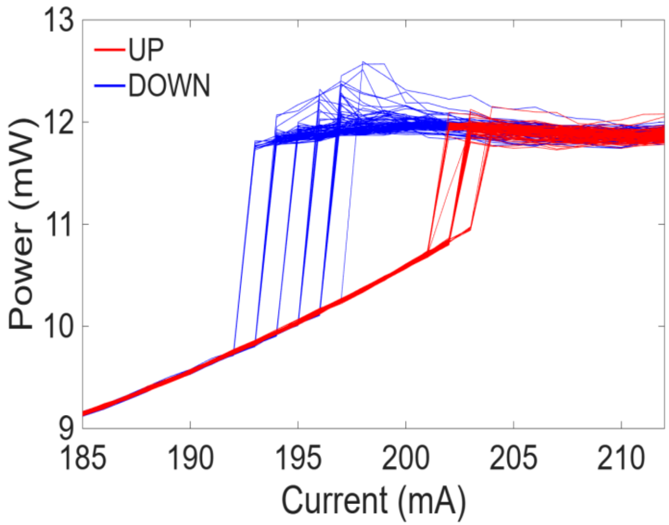








Bistability

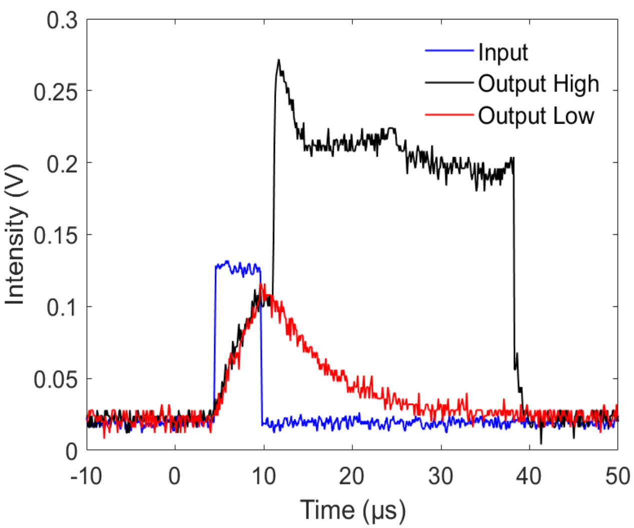


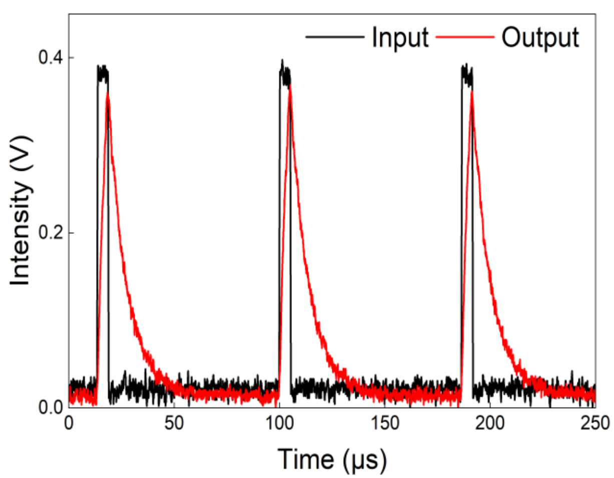
Two emission states are observed with current sweeps (upward/downward).

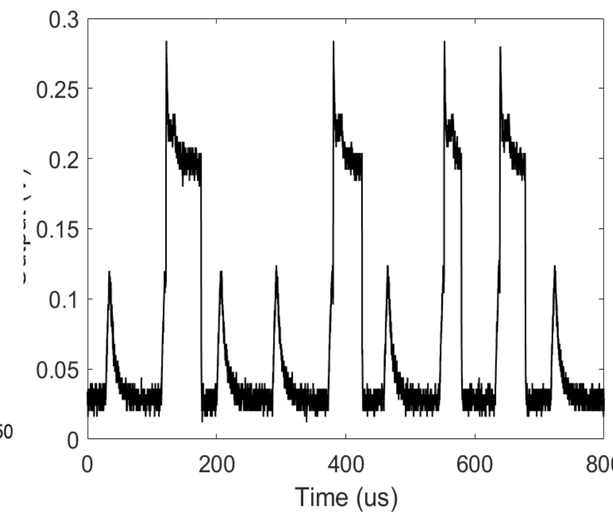
Pulse modulation → HIGH state excitation

AC modulation:

- I_{BIAS} = 195 mA
- pulse width **5 µs**
- pulse period 100 µs
- variable amplitude

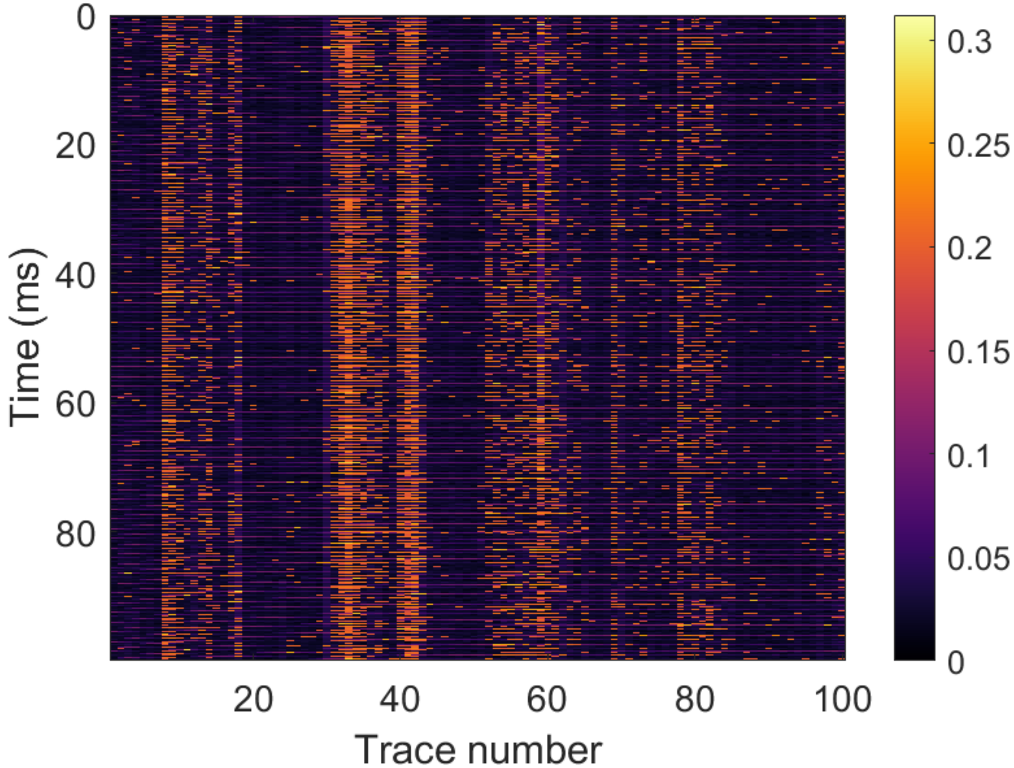


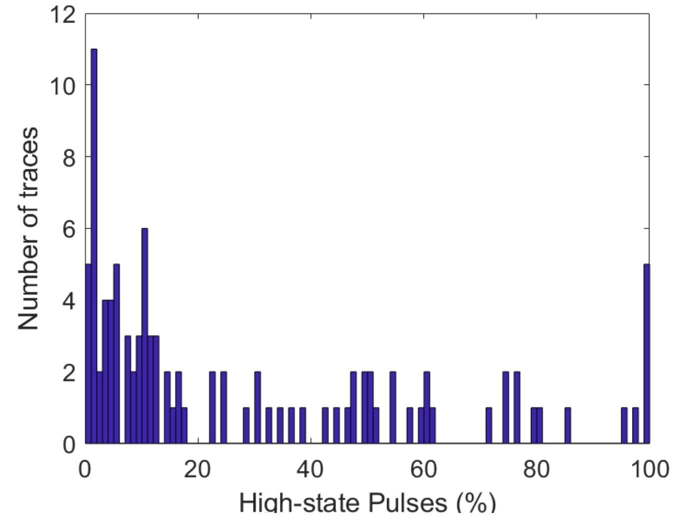


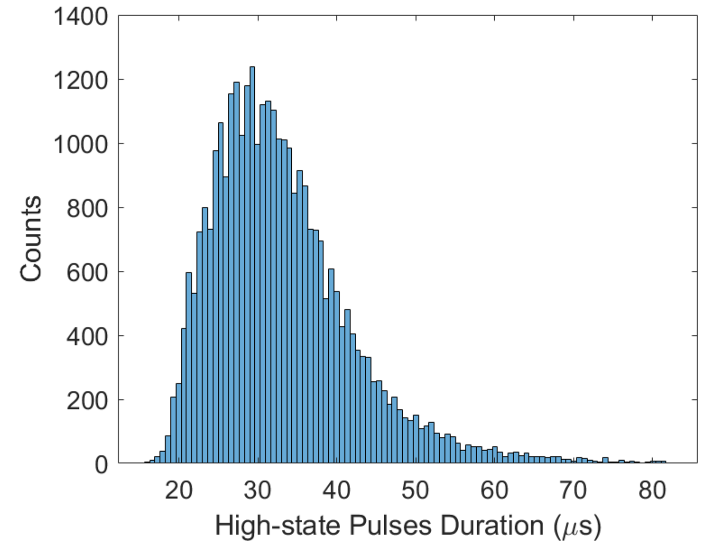


Residence time of high state

Long duration (100 ms) non-consecutive acquisitions (100).



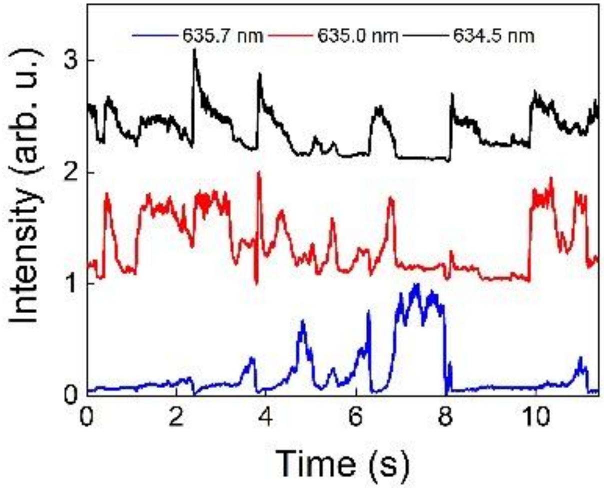


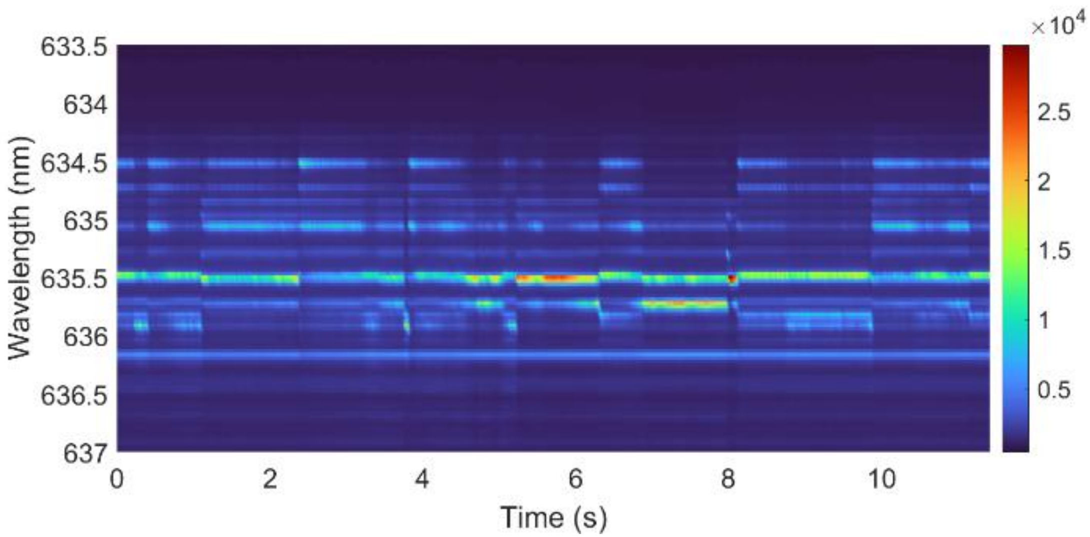


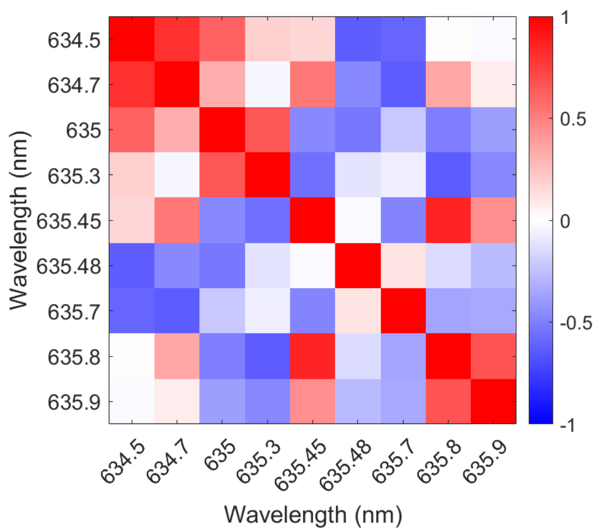
Mode competition

Constant current 245 mA, above hysteresis region.

Correlated and anti correlated stochastic spectral fluctuations.

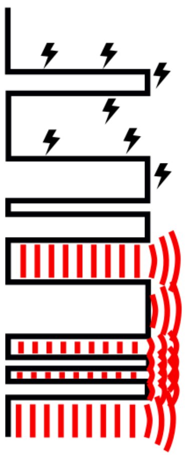






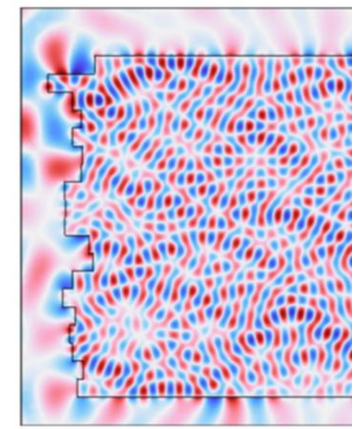
Discussion

FIB processed facet



Non-radiative recombination centers, due to dangling bonds, lattice defects and ion implantation. **Scattered optical feedback**, multimode resonance condition.

Non-radiative recombination centers induce local variations of carrier density and temperature.



Nonlinear carrier–thermal–optical feedback loop

$N(\boldsymbol{r}) \rightarrow \mathrm{T}(\boldsymbol{r}),\ n(\boldsymbol{r}) \rightarrow Feedback \rightarrow E(\boldsymbol{r})\ Modes \rightarrow N(\boldsymbol{r})$

$N(\boldsymbol{r})$ *Carriers*
$T(\boldsymbol{r})$ *Temperature*
$n(\boldsymbol{r})$ *Refractuve index*
$E(\boldsymbol{r})$ *Optical field*
$\boldsymbol{r}$ *Spatial variable*

At the milled facet, current-dependent activation of non-radiative recombination centers produces local variations of N and T, modifying the refractive index and scattered feedback, which reshape the optical field and carrier distribution.

Discussion

Multi stability

Different multimode emission states can coexist at the same current I_X, with distinct carrier, thermal, refractive-index and optical-field distributions.

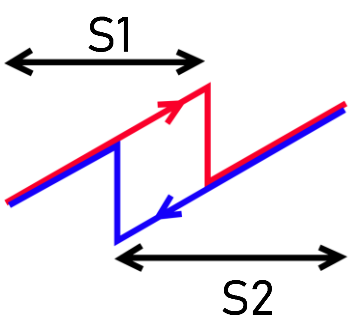
Spectral Signature 1
State 1 (S1) $N_1(\boldsymbol{r}), T_1(\boldsymbol{r}), n_1(\boldsymbol{r}), E_1(\boldsymbol{r})$

$$\Big\updownarrow \boldsymbol{I_X}$$

Spectral Signature 2
State 2 (S2) $N_2(\boldsymbol{r}), T_2(\boldsymbol{r}), n_2(\boldsymbol{r}), E_2(\boldsymbol{r})$

Hysteresis

Overlapping stability regions allow S1 and S2 to coexist at the same current. The observed state S(t) depends on the system's history through *N(t,r), T(t,r), n(t,r),* and *E(t,r).*



Spectral dynamic competition

Stochastic switching between multimode unstable states attributed to fluctuations of the coupled carrier-thermal-optical system

Summary and conclusions

Main results

- Reliable and reproducible method for RLD realization.
- Optical bistability, hysteresis, and stochastic switching of the modal configuration at constant current.
- Qualitative interpretation of the results attributed to the interplay between carrier/thermal dynamics and disordered optical feedback introduced by the milled facet.

Applications

- Single photonic platform for non-linear dynamics (no need for external optical feedback or injection)
- Signal processing, computation

Thank you

arocoba.github.io
antonio.consoli@urjc.es
antonio.consoli@csic.es